

RESEARCH PROPOSAL
BSc Hons Geoinformatics

**PREDICTIVE GEOSPATIAL MODELLING OF JOURNEY-FROM-CRIME OF
STREET ROBBERIES IN A SOUTH AFRICAN TOWN**

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1 BACKGROUND

Street robberies, characterised as a violent, predatory crime where an offender threatens or physically harms a victim in a public or semi-public space with the specific aim of stealing the victim's property (Monk et al., 2010). Street robberies are a largely ignored problem within South Africa, with the media and SAPS often focussing on other violent crimes such as high-profile robberies, property theft and cash-in-transit heists (ISS Africa, 2020). ISS Africa advocates for a strategic policing focus on street-level offences, warning that perpetrators of street robberies often escalate their criminal trajectories into highly violent, organised crimes like carjackings and home invasions if left unchecked. According to SAPS statistics in 2018, there were over 140 000 cases of aggravated and serious armed robbery between April 2018 and March 2019 with 57% of those crimes being committed in a public space (ISS Africa, 2020).

Stellenbosch, a satellite town bordering the Cape Town Municipality, has not been spared from the increasing violent crime within South Africa. This is highlighted due to Stellenbosch having seen a tripling in street robberies from 2021 to 2025 ("How safe is Stellenbosch, Western Cape?", 2025). Stellenbosch has a large student, tourist and working-class population causing it to have a large pedestrian population (*PROPOSED NETWORK & STRATEGIES NMT Masterplan*, 2020). The town's large reliance on its pedestrian population makes it particularly susceptible to the consequences of street robberies. These crimes affect the physical and psychological safety of the community while simultaneously destabilising the local economy which is reliant on tourism, nightlife and high pedestrian traffic.

Reactive methodologies have been the main methodologies utilised by SAPS and private law enforcement to combat street robberies in the past. Traditional hotspot mapping (THM) has been the primary methodology. THM is an inherently flawed method due to its heavy reliance on spatio-temporal factors. This causes it too often lack the ability to reduce crime but rather to move the crimes from one susceptible area to another (Nakaya & Yano, 2010). This is evident due to Stellenbosch's inability to significantly reduce the number of street robbery in the town despite the increased law enforcement and private security presence in the town (*POLICE RECORDED CRIME STATISTICS REPUBLIC of SOUTH AFRICA*, 2025).

A major impediment to traditional spatial modelling in South Africa is the dark figure of crime which is the substantial gap between actual crime occurrences and incidents officially recorded by police. Nationally, street robbery reporting rates sat at a critical low of 44% (Maluleke, 2024), due to public distrust toward the South African Police Service (Maluleke, 2023). This statistical unreliability is compounded by administrative zonation effects and data aggregation errors, where SAPS data points are routinely offset to the nearest street intersection to protect anonymity, therefore separating the crime data from the micro-environmental features that may have facilitated the offence.

Recent advances in GIT have created a methodology that has the ability to bypass the flawed offence data by focussing on the built environment around the crimes rather than where the crimes cluster. Risk Terrain Modelling (RTM) is a spatial analytical technique used to identify how environmental and landscape features combine to create high-risk locations for specific events (Buccine-Schraeder & Kennedy, 2021). These features are often municipal infrastructure such as lighting, transport nodes or land use types. RTMs allow researchers to evaluate how the built environment (or the lack thereof) co-locate to create high or low risk areas. RTMs have shown to be better at predicting where crimes occur than THM in both

Western and South African contexts. However, current literature shows that Western-centric environmental weightings often fall short when applied to South Africa's unique urban characteristics. South African research indicates spatial paradoxes where Western assumptions of safety do not correlate within South African landscapes. To solve this limitation, this study will exclusively use South African spatial literature to derive localized environmental risk weightings, ensuring the baseline RTM accurately reflects the reality of local urban dynamics.

Traditional spatial routing has explored an offender's journey to a crime, post-offense routing (Journey-from-crime) has had very little research done focusing on it in Western literature and is a gap in South African literature. To address this universal gap, this study uses Routine Activity Theory (Cohen & Felson, 1979), Crime Pattern Theory, and the Rational Choice Perspective to simulate how an offender escapes.

By treating the risk landscape as a cost surface layer, the study will operationalize diverse Offender Archetypes (Stealth-Oriented, Speed-Oriented, and Balanced-Utility). These behavioral profiles will be simulated using cost accumulation and routing algorithms, specifically modified Dijkstra's shortest-path logic and paired with stochastic Monte Carlo simulations to account for bounded rationality and micro-level data uncertainty.

2 PROBLEM STATEMENT

While street robberies are escalating rapidly, current security and policing responses are hindered by a lack of predictive, data-centric insights into offender movements post-incident. Although Risk Terrain Modelling (RTM) successfully predicts the likelihood of crime occurrences (victimisation), there is a significant theoretical and practical gap in applying these models to predict *post-crime* escape routes (Journey-From-Crime) within a South African context. Furthermore, RTMs have not been extensively built in a South African environment and have never been built for informal townships and towns.

Existing spatial routing methodologies struggle to simultaneously account for the various complex priorities of fleeing offenders; specifically, the balance between rational risk-aversion (stealth), time-minimization (speed), and opportunistic, stochastic human behaviour over raster surfaces. Consequently, there is an inability to accurately simulate and aggregate these offender archetypes to identify chokepoints with the environment, which are could assist greatly in optimising the placement of security infrastructure and rapid police deployments.

Two RTMs will be developed for this study, one will define risk as a risk of a street robbery occurring, while the second will define the risk as the likelihood of a fleeing robber choosing a particular route to escape post robbery. The model will be based purely on literature-derived risk terrain models and tested against real-world incident data to assess its predictive accuracy.

2.1 Research Questions

- How do different offender archetypes alter their spatial routing through the Escape Terrain Model (ETM) based on competing priorities of risk-aversion, time, and stochastic unpredictability?

- Where are the primary geospatial chokepoints located when aggregating the simulated escape routes into a single probability density surface?
- To what extent does the literature-based Crime Risk Model accurately predict real-world spatial crime distributions when validated against historical SAPS incident data?

3 AIM AND OBJECTIVES

This study aims to predict how offenders are likely to move immediately after a street robbery, identifying chokepoints where infrastructure and policing can most effectively deter offending actions and improve rapid response.

The following objectives have been set to achieve the research aim:

- a) Objective 1. Conduct a literature review to determine the most appropriate South African environmental risk weightings and spatial routing algorithms for modelling post-robbery escape routes.
- b) Objective 2. Collect and prepare spatial data of Stellenbosch's environmental features alongside historical SAPS street robbery records to serve as the foundation for the models.
- c) Objective 3. Collect, clean and prepare infrastructure and land use types in Stellenbosch.
- d) Objective 4. Create an RTM of the risk of the street-robbery victimisation.
- e) Objective 5. Create an RTM cost raster of the likelihood of a fleeing offender choosing a particular street/cell for their escape.
- f) Objective 6. Simulate the escape paths of three different offender archetypes using routing algorithms to pinpoint critical environmental chokepoints.
- g) Objective 7. Assess the accuracy and predictive power of the first RTM using historical street robbery data supplied by SAPS.

4 SIGNIFICANCE AND RATIONALE

Street robberies pose a threat to the safety and economic stability of South African urban centres and towns. Traditional policing and existing spatial models focus on where crimes

are likely to occur or cluster (victimisation) and how the offenders get to their crimes (Journey-to-Crime) but lack the ability to predict the way offenders move post-crime (Journey-from-Crime). Reliance on the international, Western-centric environmental weightings to model crime often fails to accurately reflect South Africa's unique urban morphology. Furthermore, while existing Risk Terrain Models built within a South African context have predominantly focused on major metropolitan areas such as Johannesburg, Pretoria, and Cape Town, approximately 30% of the national population resides in smaller towns (Ritchie, 2026). This shows that there is a need to develop and validate RTMs specifically tailored to a small South African town setting. Consequently, current security responses (especially in smaller towns) are hindered by a lack of predictive, localized spatial data regarding offender movements.

The current aim and objectives are fitted to address the research gaps identified as well as the methodological limitations identified. The study will use an Analytical Hierarchy Process (AHP), to derive literature-based weightings specific to a South African context (Objective 1). The model will be validated using empirical private security and SAPS spatial crime data (Objectives 2 and 7). This allows the research to avoid the limitation introduced by using Western literature. These steps will assist in the creation and validation of the Crime Risk Model (Objectives 4 and 7). The development of the Escape Terrain Model (Objective 4) will utilize the collected infrastructure data (Objective 2) to directly address the journey-from-crime gap identified in the literature. Finally, by using 3 different routing algorithms, the three offender archetypes will be modelled and chokepoints in the environment will be identified (Objective 6). The 3 different routing algorithms will create a very robust model directly addressing the highly unpredictable nature of the offender movements.

This study has the potential to have both a real-world actionable and academic impact. Academically, the study creates a methodological framework for simulating the highly under-researched journey-from-crime, advancing the application of Risk Terrain Modelling in South Africa and potentially in other developing nations. It also has the potential for a real-world impact by informing SAPS, Law enforcement and private security companies of high-probability escape routes and spatial chokepoints. This information can assist in data-driven decision making by these stakeholders on where infrastructure, patrol routes and other crime prevention strategies should be placed and the budget that should be allocated to each of these strategies. This data driven approach could optimise their budgets and improve the effectiveness of their crime prevention strategies.

5 RESEARCH METHODOLOGY

5.1 The Approach

This study adopts a Positivist paradigm, as it assumes that spatial offender behaviour can be objectively measured and modelled. The research is predictive and model-building in nature, aiming to use established theoretical links (Routine Activity Theory, Rational Choice Perspective and Crime Pattern Theory) to forecast post-crime movement. It utilizes a deductive reasoning approach, moving from established criminological theories to testing these concepts against simulated and real-world spatial data.

5.2 The Data Types

The study relies strictly on secondary, quantitative spatial data. This includes vector and raster datasets for environmental features, and historical point data for crime incidents.

5.3 Method Types

A quantitative, computational spatial analysis approach is used. This involves spatial multicriteria evaluation developing RTMs, deterministic routing algorithms, and stochastic probability simulations.

Chapter 1 - Introduction

- Real-world problem of street robbery displacement, outlines the study area

Chapter 2 - Literature Review

- Explanation of Environmental Criminology theories, spatial risk forecasting, and the existing gaps regarding the journey-from-crime.

Chapter 3 - Methodology

- Details the specific secondary datasets utilized, the data preparation processes, and the mathematical mechanics of the spatial algorithms

Chapter 4 - Results & Analysis

- Presents the outputs of the spatial multicriteria evaluation, mapping the localized environmental weightings, and visualizes the simulated post-offence escape routes and identified spatial chokepoints

Chapter 5 - Discussion & Conclusion

- Evaluates the success of the predictive models against historical data, contextualizes the findings within the South African urban landscape, and provides actionable recommendations for infrastructural and policing deployments.

Figure 1: The structure of the study.

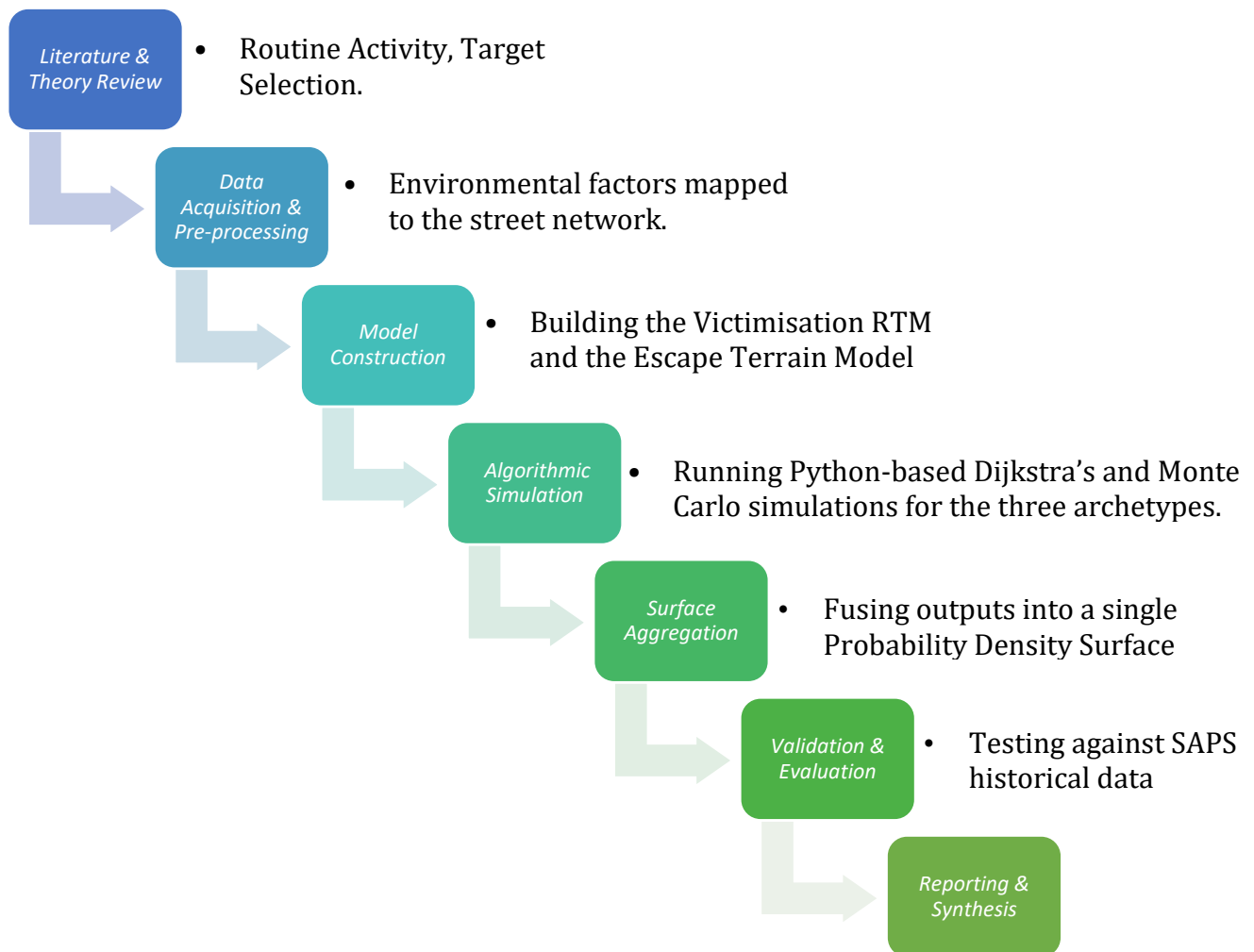


Figure 2: The Study Agenda

6 RESEARCH METHODS

6.1 Study area

Stellenbosch is a satellite town that borders the Cape Town municipality in South Africa. It has a large student population causing it to have a high pedestrian traffic. This high pedestrian traffic makes the town more susceptible to street robbery (Yue, Liu & Xiao, 2023). The town has very heterogeneous landcover uses such as clubs and bars, university ground and student accommodation and public transport pickup spots. This heterogeneity allows for easier transferability to other South African towns and cities. It has a medium population size of 200 000 residents which makes it suitable for the models' transferability to small cities such as Bloemfontein or small towns such as Knysna. It was also chosen because it will

make an impact within the Stellenbosch Community which surrounds Stellenbosch University.

6.2 Data

Due to the unreliability of crime data within South Africa (Hollands, 2024), no real crime incident data will be used in the creation of the RTM. Both RTMs will be created using environmental factors which the weights and spatial influence buffers are informed by literature. This literature-based approach is suitable for scenario testing.

6.2.1 Environmental Factors

Pedestrian Traffic Generators such as bars and clubs, ATMs, banks, shops and malls, student residences, campus entrances, and parks will be used (Brantingham & Brantingham, 1993). Transport points such as bus stops, taxi ranks and parking areas are also included within the model (Brantingham & Brantingham, 1993; Savard, 2018; Bailey & Loggenberg, 2023). Streetlamps and lighting networks will also be used (Savard, 2018). Spatial influence (proximity/density) and grid size follow RTMDx guidance (Marchment & Gill, 2021). Feature weights begin from reported RRVs/effects in the literature (Caplan et al., 2015; Marchment & Gill, 2021) and are normalised to (0,1) for routing cost/utility.

6.2.2 Risk Terrain Model

A risk terrain model is a modelling technique which analyses the features of an environment and assesses how those features combine to create high and low risk areas for crime (Thompson, n.d.). These features will be included, excluded or weighted differently depending on the time of day that the RTM is designed for, for example, during the morning, bars are often closed and therefore do not attract crime, but during the night where bars are often very busy with many inebriated people, it attracts criminals looking for easier targets. Murders are often a good indication for when and where other forms of violent crimes occur. According to a 2019 SAPS crime report, the majority of murders occurred during the hours

of 6pm and 3am, therefore our model and attractors will be based on features that are prevalent within the night and late evening environment.

6.2.3 Weightings

6.2.3.1 Risk Terrain Model 1

Risk Terrain Model 1 is hereafter referred to as the Crime Risk Model (CRM). The Crime Risk model is a traditional RTM that will determine the likelihood of a street robbery occurring any point within the study area. To achieve this, the model utilizes a network-constrained

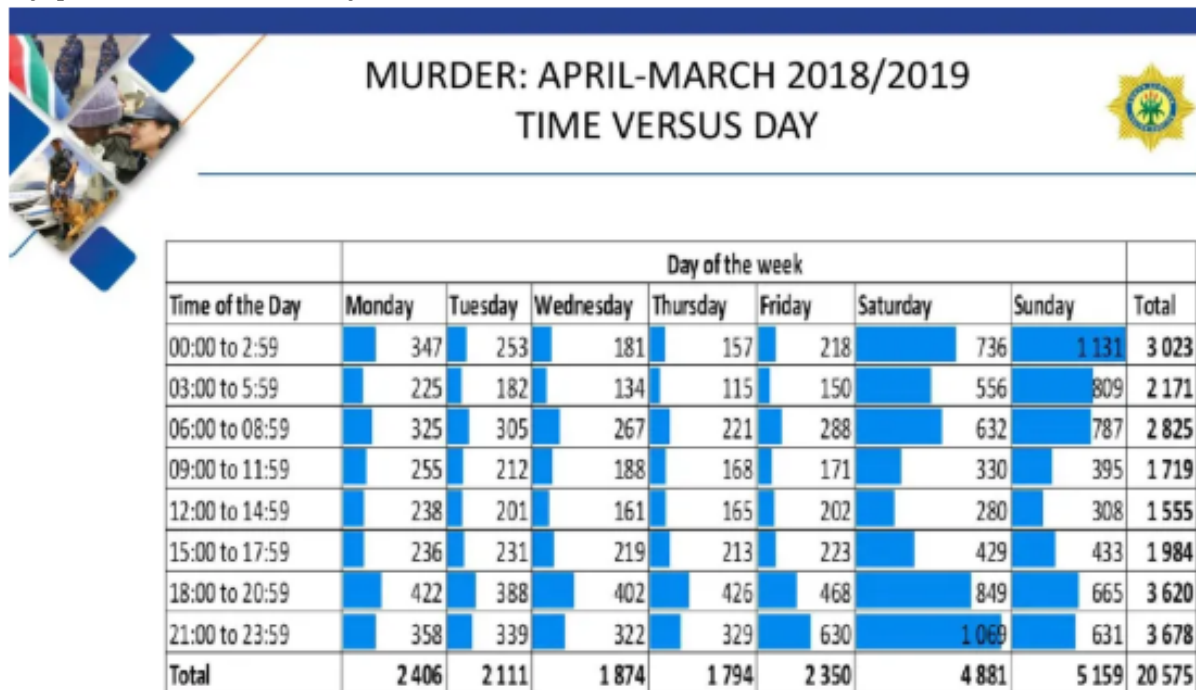


Figure 3: The number of murders per day per time frame in Cape Town.

distance accumulation approach, calculating the Manhattan distance from each attractor along the street network rather than using simple Euclidean buffers. These distances are then reclassified into a discrete risk scale ranging from 5 (highest risk) to 1 (lowest risk) and combined through a weighted sum overlay based on the specific attractive levels of each factor. The weights used in the weighted sum are purely literature derived based predict on the likely areas of where street robberies may occur. These will then be aggregated into points as the starting points for where the crimes will occur. The accuracy of this model will then be assessed using the real-world crime data.

Table 1: Theoretical Attractors & Normalised Weights (derived from Savard, 2018; Caplan et al., 2015; Brantingham & Brantingham, 1993; Marchment & Gill, 2021)

Attractor	Rationale (brief)	Prior weight (0–1)
Taxi Rank	Taxi Ranks are commonly known in South Africa for being crime hotspots	0.90
Poor/failed lighting segments	Reduced visibility; offender advantage	0.90
Nightlife nodes (bars, clubs, liquor on-sales)	Targets + intoxication; weak guardianship late	0.85
ATMs / bank frontages	Cash handling; stationary targets	0.80
Public transport stops (Bus/Train stations)	Converging flows; waiting targets	0.80
High-betweenness segments (desire lines)	Offender search & movement paths	0.70
Vacant / problem buildings	Refuge/concealment; weak informal control	0.70
Alleys / laneways / service passages	Low visibility; escape options	0.65
Parking lots / garages	From/to vehicles; limited surveillance	0.65
Land-use edges (retail↔residential; park↔residential)	Transition zones; predictable flows	0.60
Student residence perimeters / gates	Target density at arrival/departure	0.60
Parks / greenbelts	Low guardianship; refuge/escape	0.55
Retail clusters	Closing routines; fewer witnesses	0.50

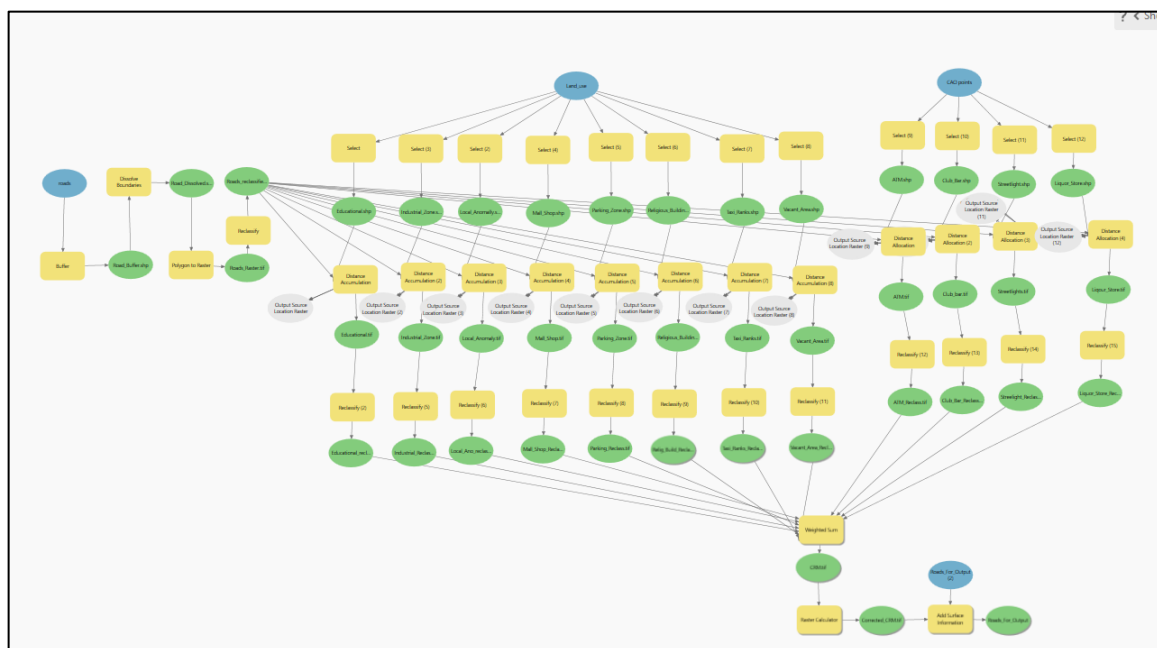


Figure 4: Workflow for the creation of the CRM

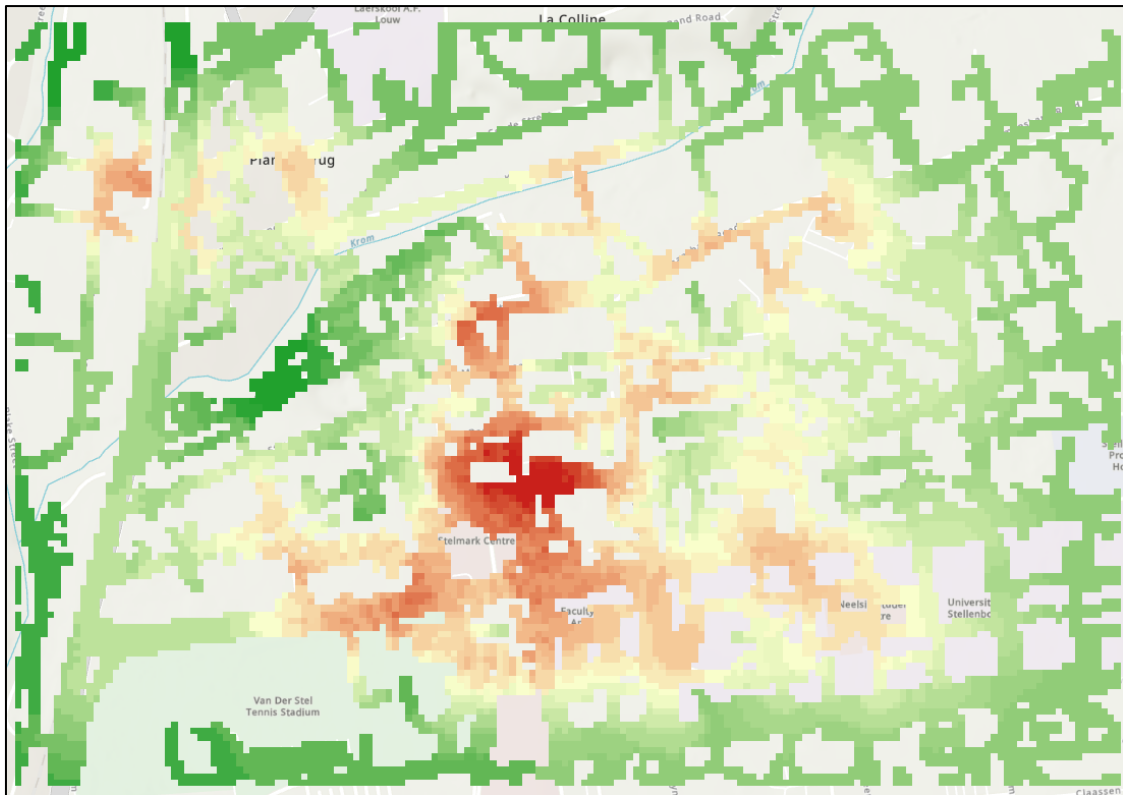


Figure 5: Risk Terrain Model, CRM, generated from the workflow above of a small area in Stellenbosch

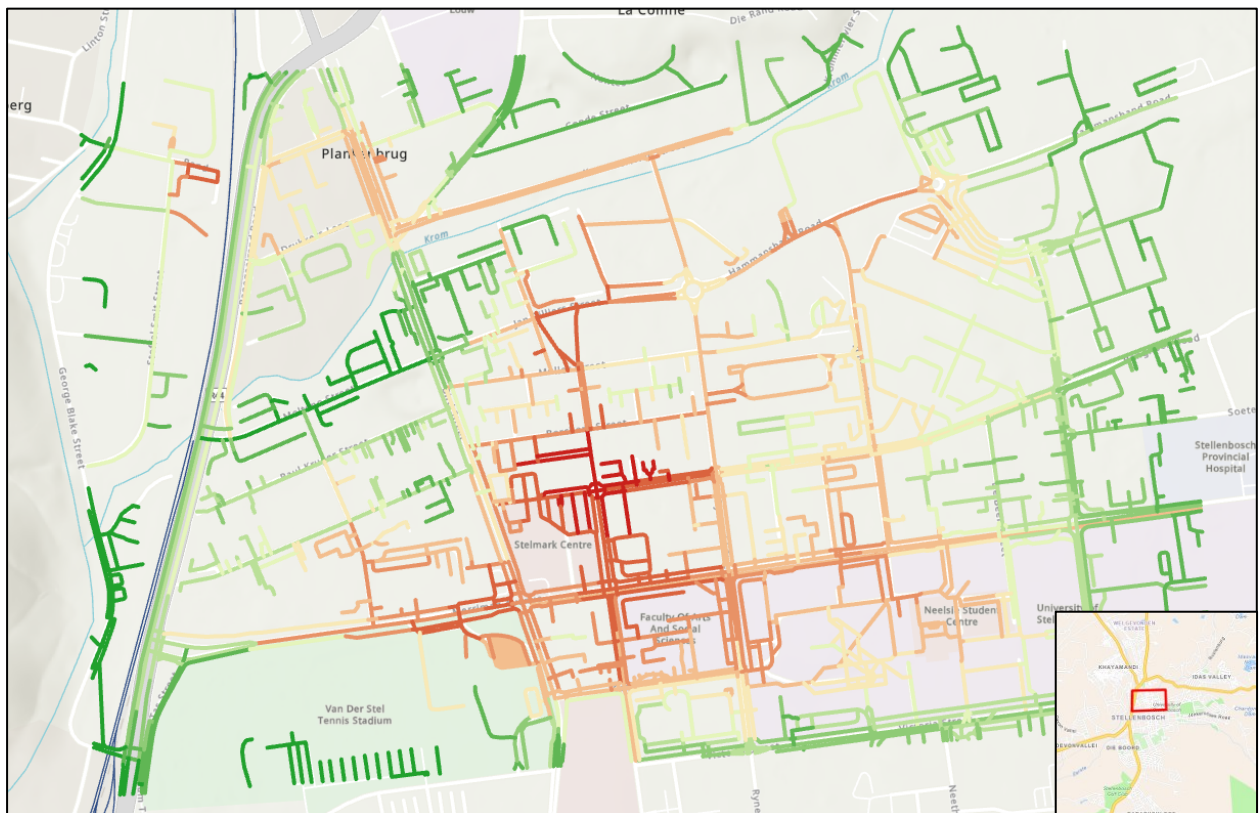


Figure 6: The CRM converted into a Street Map for Visual Interpretation

6.2.3.2 Risk Terrain Model 2

Risk Terrain Model 2 is hereafter referred to as the Escape Terrain Model (ETM). The Escape Terrain Model will determine the likelihood of a criminal using that cell during their escape post-robbery. The ETM will only be used for two of the offender archetypes namely the Stealth-Oriented and the Balanced-Oriented. These offender archetypes will be discussed in detail in 2.2. The ETM will be built using logical factors including lighting, patrol routes, high pedestrian volumes, etc. Higher values will imply a higher probability of detection / risk and therefore a less likelihood of the archetypes mentioned above choosing those cells.

Table 2: Capture/Visibility Contributors & Normalised Weights (derived from Savard, 2018; Caplan et al., 2015; Brantingham & Brantingham, 1993; Marchment & Gill, 2021)

Guardianship / Visibility	Rationale	Prior weight (0–1)	Notes
ALPR/LPR at vehicular egress	Link to vehicles; case development	0.85	Bridges, arterials, campus gates
Bright, uniform LED lighting	Natural surveillance; reduce concealment	0.85	Prioritise shadow-free, high-CRI
Active patrol presence (campus/municipal/private)	Capable guardianship; deterrence	0.85	Time-targeted to risk periods
Controlled access (gates/turnstiles)	Flow regulation; screening	0.75	Especially at campus thresholds
High pedestrian volume (eyes on street)	Informal guardianship	0.65	Time-varying; may invert at extremes
Open sightlines / trimmed vegetation	Reduce refuge; increase visibility	0.60	Pair with lighting upgrades
Monitored CCTV at decision points	Real-time guardianship; identification	0.40	Place at egresses/chokepoints, not just over nodes

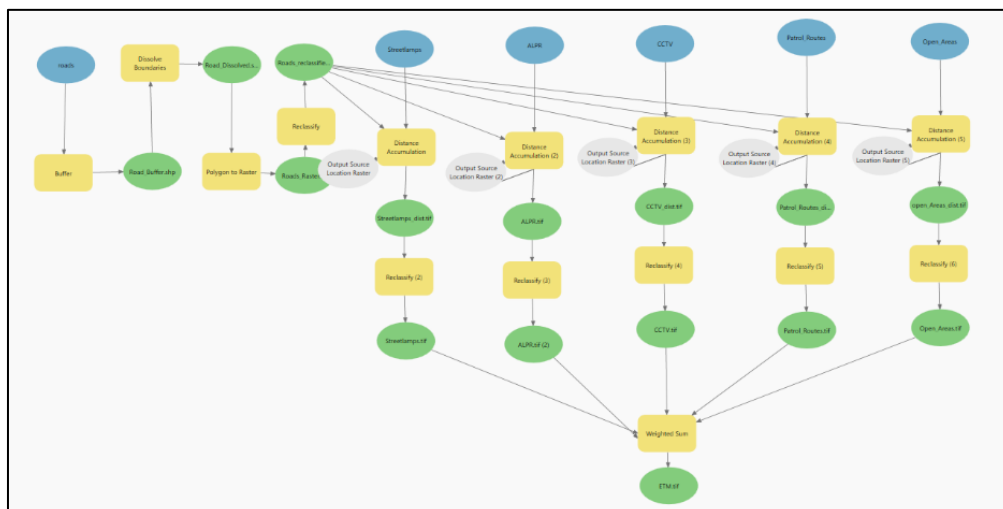


Figure 7: Workflow of the generation of the ETM

6.2.4 The Offender Archetypes

This study will conceptualise three offender archetypes, representing idealised patterns of post-crime movement behaviour in response to identical environmental conditions.

Three offender archetypes will be modelled using different routing algorithms for this study.

6.2.4.1 Environmental Factors

This archetype represents an offender that prioritises minimising their risk of being detected over the time in which it takes to get to their end destination. When modelling this offender over the ETM, an accumulation cost minimization approach will be taken.

This approach will take the form of a least cost path modelling which will use the modified Dijkstra's Algorithm below to mimic the offenders thought process. The offender will be treated as a rational agent who has extensive knowledge of their environment and its risks and has preplanned an escape route prior to committing the robbery.

$$Cost_{total} = \sum_{i=1}^n (RTM_i \times d_i)$$

Where:

n : The total number of cells or segments in the path.

i : The specific cell/segment being traversed.

RTM_i : The risk value of that specific cell (from your RTM surface).

d_i : The distance travelled across that cell

This algorithm uses very similar logic to a traditional Dijkstra's algorithm but has a few key differences. Firstly, this algorithm, unlike a traditional Dijkstra's algorithm, is designed for a raster dataset rather than a vector dataset. Secondly, this algorithm redefines the edge weights to represent the risk accumulation rather than the distance accumulation. The second difference is the most important difference as the stealth-oriented offender algorithm will treat a 10-meter stretch of high-risk terrain effectively 'further away' in cost than a 100-meter stretch of low-risk terrain. This archetype will be modelled using the distance accumulation and optimal path by line tools.

6.2.4.2 Speed Oriented Offender

This archetype represents offenders who prioritise reducing their travel time between the crime they just committed and the dumping zone destinations regardless of the detection or exposure risk. Similarly to the Stealth oriented offender, these offenders are also assumed to be rational agents who have extensive knowledge of their surroundings and therefore always choose the shortest route.

This archetype will be modelled using a traditional Dijkstra's algorithm to determine the shortest route from each crime point to each dumping zone point using the formula below.

$$Cost_{time} = \sum_{i=1}^n \left(\frac{d_i}{v_i} \right)$$

with

$\sum_{i=1}^n$ (sum): This calculates total accumulated travel time for the entire escape route

d_i (Distance): The distance travelled across the specific cell

v_i (Travel Time): The travel speed through the specific cell.

This equation ensures that the travel time is minimized by considering both the distance and speed through the terrain. This archetype will be modelled using the distance accumulation and optimal path by line tools

6.2.4.3 Balanced Utility Offender

This archetype represents an offender that prioritises neither stealth nor time. Unlike other two archetypes, the balanced-utility offender operates on a weighted balance of both the ETM and the speed model.

$$Cost_{cell} = [\omega \cdot F_{time} + (1 - \omega) \cdot R_{risk}] + \epsilon$$

With

ω (Omega): The Normalised Weighting Factor. This represents the offender's priority.

F_{time} (Time Friction): A raster where pixel values represent the impedance of the terrain

R_{risk} (Risk Impedance): The normalized values from your Escape Terrain Model (0 to 1).
 ε (Epsilon): The Stochastic Noise term (Random noise added to each cell to vary the path based on the Euclidean distance from robbery point to dumping ground).

A multi-criteria evaluation (MCE) must be conducted in order to utilise this formula. The F_{time} and R_{risk} are in two distinctly different spatial units and therefore require normalisation to the same unitless scale to ensure mathematical validity. The scale will be a continuous with values ranging from 0 to 1. The equation above can then be utilised to create a hybrid impedance surface fusing the ETM and time layers together based on the weighted omega factor. A stochastic factor is added to imitate a human's unpredictability. A Monte Carlo simulation will be run many iterations based on the number of starting crime points created from the CRM. The results are aggregated to produce a probability density surface (PDS). This surface will identify high-frequency escape routes/corridors even when accounting for offender uncertainty and non-planning. Unlike the previous two archetypes which represent rational, professional offenders, the stochastic nature of this archetype ensures that opportunistic, sub-optimally planned offenders are also accounted for within the model. This will be created using python notebooks inside of ArcGIS Pro.

The chokepoints will be identified using a probability density surface (PDS) of all three archetypes. This final PDS will be created by converting the vector line output of the stealth-oriented and speed-oriented offenders into their own PBS. The three PDS's from the three archetypes will then be fused together to form a single probability density surface.

7 TIMELINE AND BUDGET

7.1 Timeline

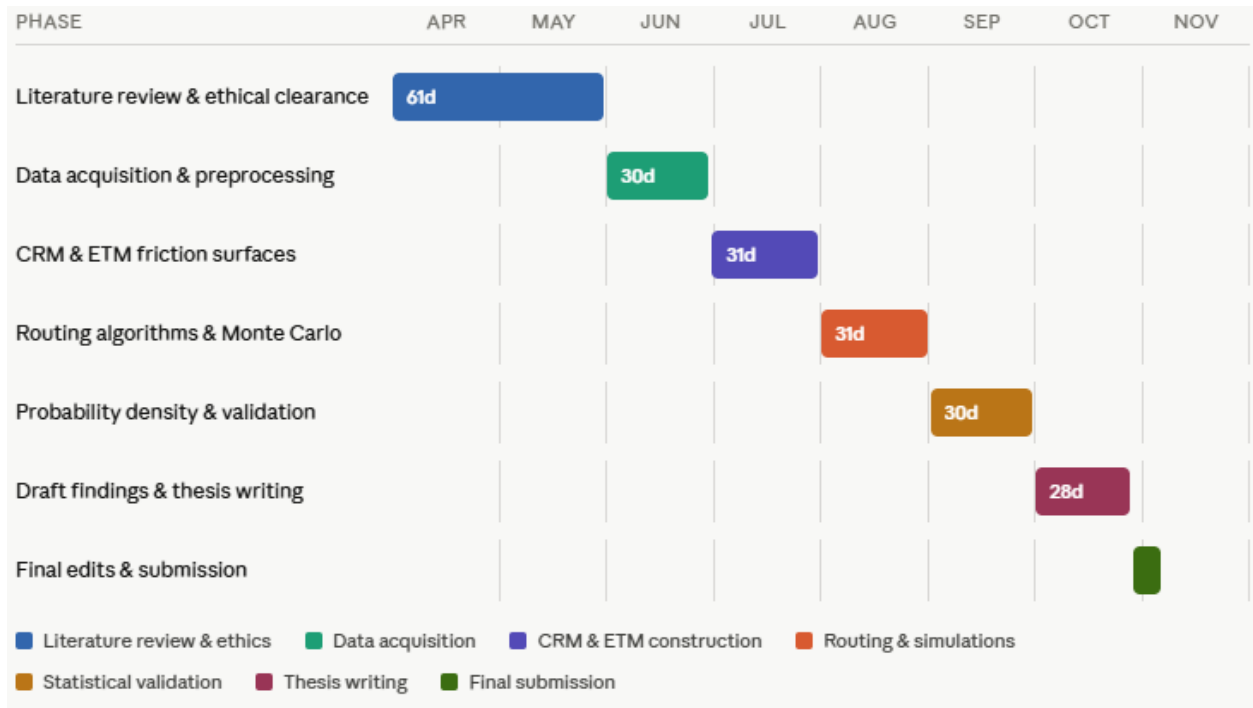


Figure 8: Gantt chart showing the timeline of the study.

7.2 Budget

- Data costs: R0.00 (Environmental data sourced from open-source platforms or municipal databases; SAPS data provided at no cost for academic research).
- Travel/fieldwork: R0.00 (Desk-based computational study; researcher is locally situated).
- Software/licenses: R0.00 (ArcGIS Pro and necessary spatial analyst extensions provided under university site license).
- Editing: R0.00 (Writing lab provided by the university).
- Encrypted Flash drive: R800 (required for sensitive data)
- Grammarly: R500 (Using for the month before submission for proofreading)

7.3 Other Considerations

7.3.1 Risks

- The primary project risk is a delay in obtaining the SAPS incident data due to bureaucratic hurdles, which could stall the validation phase.
- The data from SAPS is not detailed enough for proper validation.

7.3.2 Limitations

- **The Dark Figure of Crime:** The reliance on official SAPS data introduces inherent bias, as the low reporting rates for street robberies in South Africa mean the baseline historical data is fundamentally incomplete.
- **Spatial Aggregation Effects:** Administrative data protection practices that shift recorded crime locations to the nearest street intersection introduce spatial noise by separating the offence from its true micro-environmental context.
- **The Modifiable Areal Unit Problem (MAUP):** As raster-based spatial models, the Risk and Escape Terrain Models are inherently susceptible to MAUP, meaning their predictive outputs remain highly sensitive to the zonation effects of arbitrarily defined cell sizes.
- **Lack of Empirical Escape Data:** Because empirical micro-level data regarding an offender's exact post-crime movement is practically non-existent, the journey-from-crime model must rely on internal Monte Carlo simulations for robustness rather than real-world external validation.
- **Stochasticity of Human Behavior:** Despite utilizing established offender archetypes and algorithmic noise, computational models cannot perfectly replicate the unpredictable, bounded rationality of human spatial decision-making under severe psychological stress.

8 ETHICAL CONSIDERATIONS

- The ethical framework of this study will focus on the protection of individual privacy, complying fully with POPIA and preventing any potential harm from arising using sensitive spatial data.
- All data, including the sensitive SAPS crime data, will be stored on a password encrypted flash drive only accessible to the lead researcher and his supervisor. The flash drive will have an Apple Air tag attached to it which will be connected to the researchers' phone to ensure that the location of the flash drive is always known by

the researcher. The flash drive will remain in the Chamber of Mines, located at 82 Ryneveld Street, at all times.

- Furthermore, only street robbery and assault crime data will be used so no individual properties will be identified.
- No personal data, such as names, addresses, contact details, etc., of the victims or suspects will be used, requested or considered in the study and therefore no personal data of the victims or suspects will be published in the study.
- None of the data provided by SAPS will be shared, accessed or distributed by or to anyone other than the lead researcher and his supervisor and will only be used for the thesis proposed above.
- The actual locations of the crimes from the SAPS data will only be used to validate the model and will not be published within any public journals or shown in any presentations to ensure the privacy of the victims. Only synthetic street robbery data will be used in any public journals or presentations for illustrative purposes.
- The ETM or CRM of Stellenbosch will not be published in any public journals or shown in any public presentations to prevent criminals from using these RTMs for criminal activities. Synthetic ETM and CRM will be created for illustrative purposes for any public journals or presentations.

REFERENCES

- Bailey, C. J., & Loggenberg, K. (2023). Mapping the spatial profile and susceptibility of illegal land occupation: A case study of the Cape Town Metropolitan area, South Africa. *Abstracts of the ICA*, 6, 1–3. <https://doi.org/10.5194/ica-abs-6-12-2023>
- Brantingham, P. L., & Brantingham, P. J. (1993). Nodes, paths and edges: Considerations on the complexity of crime and the physical environment. *Journal of Environmental Psychology*, 13(1), 3–28. [https://doi.org/10.1016/s0272-4944\(05\)80212-9](https://doi.org/10.1016/s0272-4944(05)80212-9)
- Buccine-Schraeder, H., & Kennedy, L. W. (2021). Risk Terrain Modeling (RTM). *The Encyclopedia of Research Methods in Criminology and Criminal Justice*, 873–874. <https://doi.org/10.1002/9781119111931.ch169>
- Caplan, J., Kennedy, L., Barnum, J., & Piza, E. (2015). *CUNY Academic Works CUNY Academic Works Risk Terrain Modeling for Spatial Risk Assessment. Risk Terrain Modeling for Spatial Risk Assessment*.
- Cohen, L. E., & Felson, M. (1979). Social change and crime rate trends: A routine activity approach. *American Sociological Review*, 44(4), 588–608. <https://doi.org/10.2307/2094589>
- Hollands, G. (2024, October 22). *Crime statistics: who are we to believe?* GroundUp News. <https://groundup.org.za/article/crime-statistics-who-are-we-to-believe>
- How safe is Stellenbosch, Western Cape?* (2025, October 21). Es-Capetown.com. <https://es-capetown.com/en/products/blogpost/714/>
- ISSAfrica.org. (2020, January 8). *Why police should focus on street robberies*. ISS Africa. <https://issafrica.org/iss-today/why-police-should-focus-on-street-robberies>
- Maluleke, R. (2023). *GOVERNANCE, PUBLIC SAFETY AND JUSTICE SURVEY GPSJS 2022/23*. <https://www.statssa.gov.za/publications/P0340/P03402023.pdf>
- Maluleke, R. (2024). *Governance, Public Safety and Justice Survey (GPSJS) Victims of Crime 2023/24*. Stats SA.
- Marchment, Z., & Gill, P. (2021). Systematic review and meta-analysis of risk terrain modelling (RTM) as a spatial forecasting method. *Crime Science*, 10(1). <https://doi.org/10.1186/s40163-021-00149-6>

- Monk, K., Heinonen, J. A., & Eck, J. E. (2010). *Street Robbery / ASU Center for Problem-Oriented Policing*. Popcenter.asu.edu. <https://popcenter.asu.edu/content/street-robbery-0>
- Nakaya, T., & Yano, K. (2010). Visualising Crime Clusters in a Space-time Cube: An Exploratory Data-analysis Approach Using Space-time Kernel Density Estimation and Scan Statistics. *Transactions in GIS*, 14(3), 223–239. <https://doi.org/10.1111/j.1467-9671.2010.01194.x>
- POLICE RECORDED CRIME STATISTICS REPUBLIC OF SOUTH AFRICA. (2025). https://www.saps.gov.za/services/downloads/2025/2025-2026_-_3rd_Quarter_WEB.pdf
- PROPOSED NETWORK & STRATEGIES NMT Masterplan. (2020). Stellenbosch Heritage Foundation. https://www.stellenboschheritage.co.za/wp-content/uploads/NMT-Stellenbosch_Mobility-Forum-20200831_Lynne-Pretorius.pdf
- Ritchie, H. (2026, April 6). *Population and Demography*. Our World in Data. <https://ourworldindata.org/profile/population-demography/south-africa>
- Savard, D. M. (2018). A Routine Activity Approach: Assessing Victimization by Gender in Transit Environments and Other Public Locations. *Advances in Applied Sociology*, 08(01), 56–75. <https://doi.org/10.4236/aasoci.2018.81004>
- Thompson, M. (n.d.). *RISK TERRAIN MODELING: A TOOL FOR CRIME PREVENTION & REDUCTION IN NEW YORK CITY?* https://handbook.geospatial.psu.edu/sites/default/files/capstone/Thompson_Michelle_596A_report_20171101.pdf
- Yue, H., Liu, L., & Xiao, L. (2023). Investigating the effect of people on the street and streetscape physical environment on the location choice of street theft crime offenders using street view images and a discrete spatial choice model. *Applied Geography*, 157, 103025–103025. <https://doi.org/10.1016/j.apgeog.2023.103025>